Inkjet Printed QLED with Enhanced Efficiency and Stability Based on Optimized Hole Transport Layer with Less Side Emission

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Inkjet printing is an attractive technology for the fabrication of semiconductor device due to its low-cost, no-contact process and maskless on-demand patterning in contrast with other film deposition technologies. [1] inkjet printing technology has been successfully applied to electrical circuits, detectors and wearable electronic devices. [2] Since Jabbour et al. have firstly reported the fabrication of quantum dot light-emitting diode (QLED) by inkjet printing in 2009, [3] the efficiency and stability of QLED by inkjet printing have been improved. However, the ink components would erose the previous layers during the inkjet printing process and resulted in limited efficiency and operation lifetime.

Based on a classic QLED structure with poly(3,4-ethylenedioxythiophene) polystyrene sulfonate (PEDOT:PSS) as the hole injection layer, Poly[(9,9-dioctylfluorenyl-2,7-diyl-alt-(4,4’-(N-(4-butylphenyl) (TFB) as the hole transport layer (HTL) and ZnO as the electron transport layer (ETL). Also, we used cyclohexylbenzene (CHB) and decane with a ratio of 10:1 as the QD ink. The emission spectra of QD should be a single and sharp peak located at 630 nm. However, the emission spectrum of inkjet printed device with TFB annealed at 130 ℃ (which is adopted from spin-coating technology) exhibited a weak but wide side emission from 430 nm to 550 nm, as is shown in Fig. 1. This side emission is attributed to the emission spectrum of TFB. This means there are carriers leaked and recombined in TFB layer. This would definitely decrease the overall efficiency and operation lifetime of the QLED. By increasing the annealing temperature of TFB, we found that the side emission of TFB from 430 nm to 550 nm was greatly suppressed. Also, as is shown in Fig. 2. (a) the external efficiency (EQE) and luminance are both enhanced for inkjet printed device with TFB annealed at higher temperature. The peak EQE was enhanced from 0.18% to 7.52% and the maximum luminance was also enhanced from 446 cd/m^2 to 19669 cd/m^2, respectively. More importantly, the T_{50} lifetime of the inkjet printed device with L_0 at 1000 cd/cm^2 is enhance nearly 5 folds from 26 hours to 127 hours when applying higher annealing temperature of TFB, as is shown in Fig. 2. (b) The enhancement of EQE and lifetime should be attributed to the less erosion and better film quality of QD on TFB after increased the annealing temperature of TFB. Reduced side emission and pure, sharp QD emission spectrum shown in Fig. 1 could support this. The inset showed the capability of maskless patterning of QLED using inkjet printing technology.

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References

[1] Peng, X.; J. Yuan; S. Shen; M. Gao; A. S. R. Chesman; H. Yin; J. Cheng; Q. Zhang and D. Adv. Funct. Mater. 2017, 27, 1703704.

[2] Jia, S.; G. Li; P. Liu; R. Cai; H. Tang; B. Xu; Z. Wang; Z. Wu; K. Wang and X. W. Sun Adv. Funct. Mater. 2020, 30, 1910817.

[3] Haverinen, H. M.; R. A. Myllyla and G. E. Jabbour Appl. Phys. Lett. 2009, 94, 073108.